

AKSHAR AI

SUMMER INTERNSHIP REPORT

WEEK 8 | 11 JULY 2026 – 17 JULY 2026

Project Title: Pneumonia Detection System & Human Behavior Tracking System

INTERN DETAILS

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Report Period: 11 July 2026 – 17 July 2026 (Week 8)

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Project Title: Pneumonia Detection System / Hand & Gesture Recognition

1. Introduction

Week 8 was the final week of the Summer Internship program at Akshar AI Ltd. Over the previous weeks, the project moved through research, dataset preparation, model training, and web integration. The focus for this concluding week was on finalizing the technical documentation, conducting User Acceptance Testing (UAT), preparing the final presentation, and formally handing over the project.

Clear technical documentation is essential for ensuring that software can be maintained and operated by other team members in the future. A significant portion of this week was spent writing README files and detailing the API specifications for the Pneumonia Detection Web Application and the Hand Tracking module.

Additionally, User Acceptance Testing was conducted to observe how new users interacted with the application. This provided valuable feedback on the usability of the interface. Finally, the week concluded with a presentation summarizing the work completed during the internship and formally transferring the codebase to the organization.

This report summarizes the activities of Week 8, including the testing feedback, the structure of the documentation, and the final project handover processes.

2. Weekly Objectives

The objectives for Week 8 were primarily focused on documentation, testing, and communication, as the main development tasks were completed in Week 7.

The primary objectives were as follows:

1. **User Acceptance Testing (UAT):** Conduct testing sessions with team members to observe how they interact with the web interface and identify any usability issues.
2. **System Documentation:** Write a comprehensive README.md file detailing the setup process, dependencies, and instructions for running the application servers.
3. **API Documentation:** Outline the specifications for the Flask /predict endpoint, including request formats and expected JSON responses.
4. **Final Presentation:** Prepare a presentation slide deck summarizing the project timeline, methodologies, and final outcomes.
5. **Project Handover:** Ensure the codebase is clean, finalize all repository updates, and hand over the project materials to the mentors.
6. **Codebase Freeze:** Lock all dependency versions to ensure the project remains stable in the future.

3. Technical Implementation Details (Documentation)

During Week 8, the focus shifted from writing code to writing documentation.

Repository Documentation

The main repository documentation was organized to be straightforward for future developers. The README.md included:

1. **Project Overview:** A brief summary of the Pneumonia Detection and Hand Tracking features.
2. **Prerequisites:** System requirements such as Python and Git.
3. **Setup Instructions:** Commands needed to set up a virtual environment and install the required packages.
4. **Usage Guide:** Instructions on how to start the Flask backend and run the tracking scripts.

Writing documentation from the perspective of a new developer ensures that all necessary, often assumed, setup steps are explicitly stated.

4. Technical Implementation Details (API Specifications)

The /predict endpoint for the web application was documented to clarify how other applications or developers could interact with it.

The API documentation covered:

- **Request Type:** HTTP POST.
- **Content Format:** multipart/form-data.
- **Parameters:** A single file parameter expected to contain an image object (JPEG or PNG).
- **Responses:** Examples of both successful JSON payloads (containing the prediction string and a confidence float) and standard error payloads.

This level of detail ensures that the backend is predictable and can be easily maintained or expanded upon by other engineering teams within the organization.

5. Technical Implementation Details (Code Freeze)

A critical part of the handover process is the codebase freeze. This ensures the code delivered today will still execute a year from now.

Dependency Management: I utilized the pip freeze command to lock the exact versions of every Python package used during development (e.g., TensorFlow 2.14, Flask 3.0, OpenCV 4.8). This list was saved to requirements.txt.

By strictly versioning the dependencies, future developers are protected from breaking changes that often occur when libraries update their APIs. The codebase was then reviewed to ensure all debugging statements were removed before the final commit.

6. Day-wise Work Summary: Day 1 (Monday)

Focus: User Acceptance Testing

The week began with brief UAT sessions. I asked a few colleagues to test the web application by uploading images. The goal was to observe their interactions without providing any prior instructions.

The feedback was generally positive regarding the layout and usability. The drag-and-drop feature was well-received. One minor note was that an error message could be rephrased to be less technical. Instead of a standard HTTP error phrase, it was updated to clearly state: "Please upload a valid JPG or PNG image." This change was implemented immediately.

7. Day-wise Work Summary: Day 2 (Tuesday)

Focus: Writing Installation Guides

Tuesday was dedicated to writing the main README.md file. I focused on making the environment setup instructions as clear as possible.

To ensure accuracy, I tested the setup steps on a separate environment. I created a new virtual workspace, cloned the repository, and followed my own written instructions. This process confirmed that the project would run smoothly for a new user and that no critical installation steps were missed.

8. Day-wise Work Summary: Day 3 (Wednesday)

Focus: API and System Documentation

On Wednesday, I completed the technical documentation for the backend API, specifying the request and response formats.

I also created a simple flowchart to visualize how data moves between the frontend interface, the Flask server, and the machine learning model. This visual documentation will serve as a helpful reference for the team, illustrating the overall architecture at a glance.

9. Day-wise Work Summary: Day 4 (Thursday)

Focus: Presentation Preparation

Thursday was spent creating a slide presentation to summarize the internship. The presentation outlined the project's progression over the 8 weeks.

The slides covered the initial research phase, the dataset preparation, the process of scaling the model to prevent overfitting, and finally, the web application deployment. It also highlighted the core functionalities of the Hand Tracking module, ensuring a comprehensive overview of the entire internship experience.

10. Day-wise Work Summary: Day 5 (Friday)

Focus: Project Handover

Friday involved the final preparations for project handover. I ensured all local changes were pushed to the main repository.

All relevant files, including the trained model weights, source code, and documentation, were organized. The codebase was considered finalized and placed under a code freeze, meaning it was officially ready for the team to take over and no further modifications would be made on my end.

11. Day-wise Work Summary: Day 6 (Saturday)

Focus: Final Review

On the last day, I delivered the presentation to the mentors, summarizing the work completed over the 8-week period.

We reviewed the final applications and discussed the practical skills and technical concepts learned during the internship. The presentation was well-received, and the program concluded successfully with a final exit interview reflecting on the progress made.

12. Final Deliverables

The following items represent the finalized assets delivered to the organization:

- **Pneumonia Detection Web Application:** Fully functional frontend and Flask backend.
- **Trained Model:** MobileNetV2 .h5 file trained on the 1000-image dataset.
- **Gesture Tracking Module:** Real-time MediaPipe Python scripts.
- **Documentation:** README files, API specs, and system diagrams.
- **Weekly Reports:** Comprehensive logs covering the 8-week period.

13. Internship Learning Outcomes

The internship provided valuable experience across the software development lifecycle:

- **Deep Learning Application:** Transitioning models from theoretical notebooks to practical, scalable software.
- **Full-Stack Integration:** Connecting web interfaces to backend APIs asynchronously.
- **Professional Engineering Standards:** Learning the importance of thorough documentation, codebase hygiene, and user acceptance testing.
- **Technical Communication:** Presenting complex technical concepts to broader audiences.

14. Conclusion

Week 8 provided a structured conclusion to the Summer Internship at Akshar AI. By focusing on documentation and testing, the project was successfully transitioned from a development phase to a completed state that can be utilized and maintained by the organization.

The internship provided valuable experience across the software development lifecycle, from initial research and model training to full-stack integration and deployment. The skills gained in building practical, user-focused applications will be highly beneficial for future technical projects.

15. Appendix and References

- **Markdown Guide:** Used for formatting the final repository documentation.
- **Python Checklists:** Best practices utilized for freezing environments and deploying Python applications.
- **UI/UX Principles:** Applied to the final User Acceptance Testing review.